

Application to ICT

Introduction to Computers (LAB)

Week 1

Table of Content

- How to Check System(PC) Specifications
- Introduction to Computers
- Identify Computer Components
- Explore Storage Devices
- Capacity and Speed Considerations
- Steps to Install a Hard Drive

How to Check System(PC) Specifications

Basic Task for System Specification Knowledge

Task Manager

Processes

Performance

App history

Startup apps

Users

Details

Services

Type a name, publisher, or PID to search

Processes

Run new task

End task

Efficiency mode

Name	Status	26% CPU	62% Memory	0% Disk	2% Network
Apps (4)					
> Windows Explorer		1.4%	211.5 MB	0 MB/s	0 Mbps
> Task Manager		3.9%	78.3 MB	0 MB/s	0 Mbps
> Microsoft PowerPoint		0%	163.3 MB	0 MB/s	0 Mbps
> Google Chrome (17)	Efficiency ...	0%	1,617.7 MB	0.1 MB/s	0 Mbps
Background processes (115)					
> 腾讯手游助手 (32 bit)		0%	1.5 MB	0 MB/s	0 Mbps
> ZapyaService, for zapya hotsp...	Suspended	14.1%	10.2 MB	0 MB/s	0 Mbps
wmpf_installer (32 bit)		0%	1.8 MB	0 MB/s	0 Mbps
WMI Provider Host		0%	24.9 MB	0 MB/s	0 Mbps
WMI Provider Host		0%	16.4 MB	0 MB/s	0 Mbps
Windows Wireless LAN 802.11...		0%	0.9 MB	0 MB/s	0 Mbps
> Windows Widgets (8)	Efficiency ...	0%	19.1 MB	0 MB/s	0 Mbps
> Windows Shell Experience Ho...	Suspended	0%	5.3 MB	0 MB/s	0 Mbps
Windows Security notification ...		0%	1.6 MB	0 MB/s	0 Mbps
> Windows Security Health Servi...		0%	6.7 MB	0 MB/s	0 Mbps
Windows Driver Foundation - ...	Efficiency ...	0%	24.4 MB	0 MB/s	0 Mbps
> Windows Default Lock Screen ...	Suspended	0%	7.7 MB	0 MB/s	0 Mbps



Processes



Performance



App history



Startup apps



Users



Details



Services



Settings

Performance

Run new task



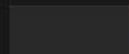
CPU

27% 3.08 GHz

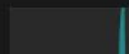


Memory

9.9/15.9 GB (62%)



Disk 0 (C:)

SSD
0%

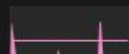
Disk 1 (D: P: E: S: H: G:)

HDD
0%

Ethernet

VMware Network Adapter V..
S: 0 R: 0 Kbps

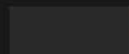
Ethernet

VMware Network Adapter V..
S: 0 R: 0 Kbps

WiFi

Wi-Fi
S: 0 R: 0 Kbps

GPU 0

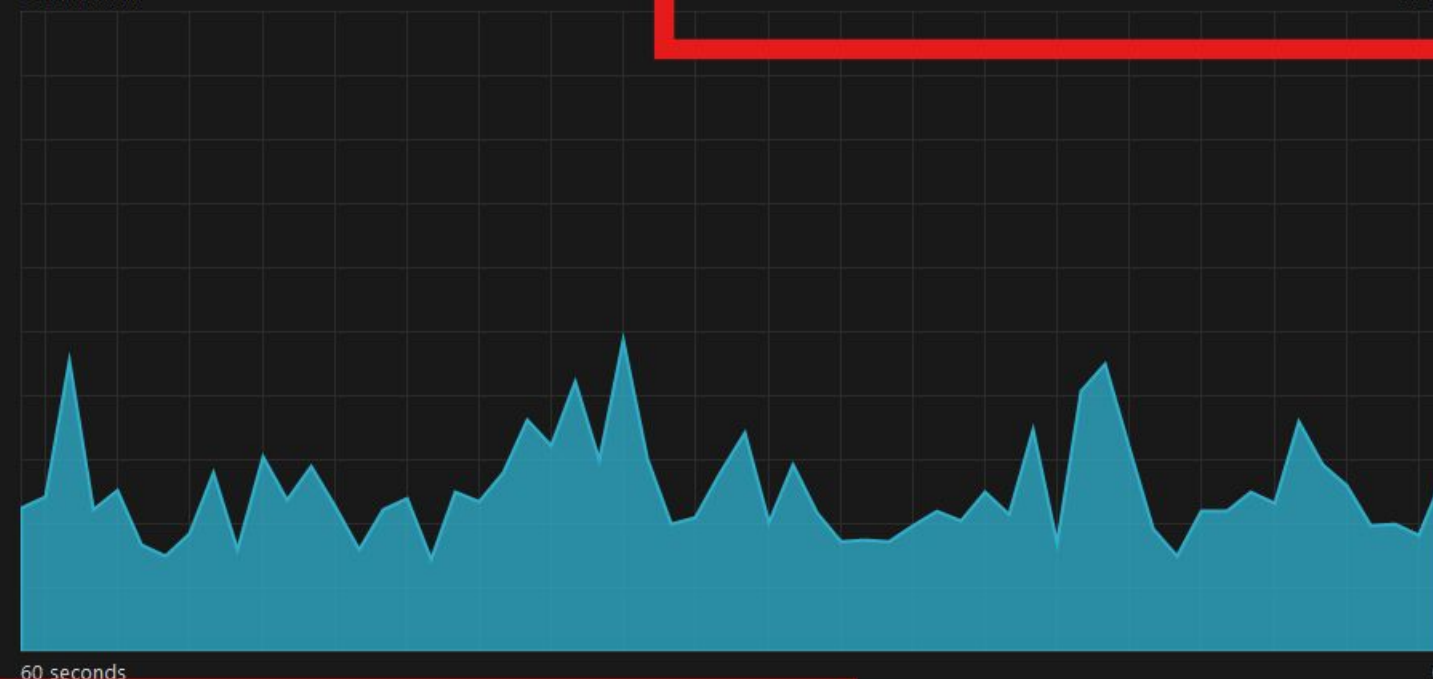
Intel(R) HD Graphics 530
10%

GPU 1

NVIDIA GeForce GTX 960M

CPU

% Utilisation



60 seconds

0

Intel(R) Core(TM) i7-6700HQ CPU @ 2.60GHz

100%

Utilisation	Speed	Base speed:	2.60 GHz
27%	3.08 GHz	Sockets:	1
Processes	Threads	Cores:	4
282	3784	Logical processors:	8
Up time	Handles	Virtualisation:	Enabled
0:09:55:50	134497	L1 cache:	256 KB
		L2 cache:	1.0 MB
		L3 cache:	6.0 MB



Processes



Performance



App history



Startup apps



Users



Details

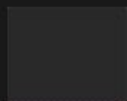
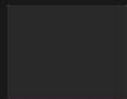
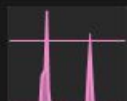
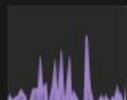
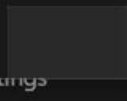


Services



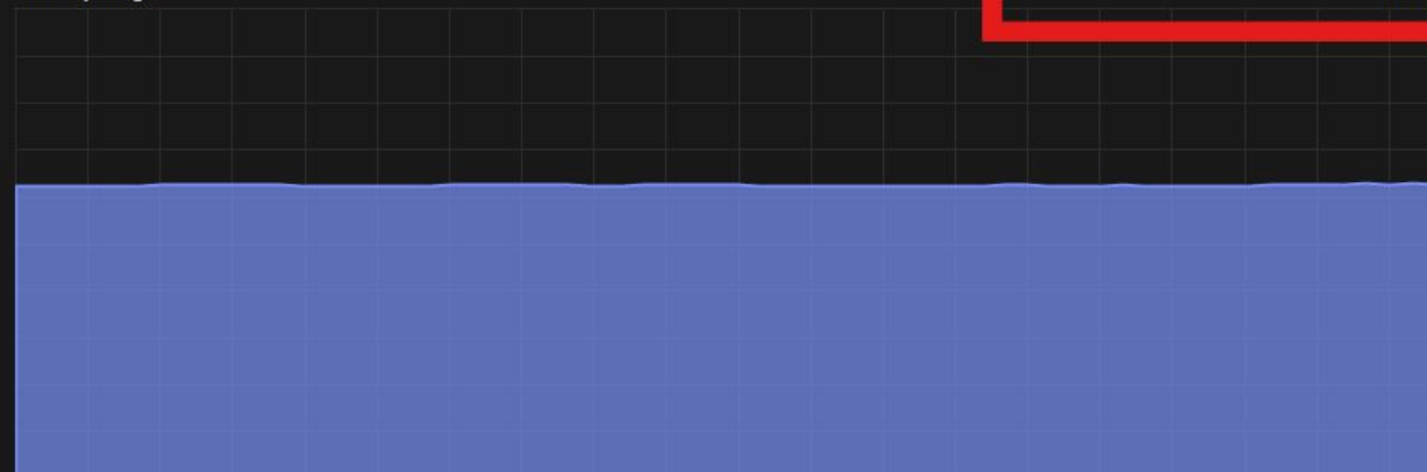
Settings

Performance

**CPU**
33% 3.07 GHz**Memory**
10.0/15.9 GB (63%)**Disk 0 (C:)**
SSD
1%**Disk 1 (D: P: E: S: H: G:)**
HDD
0%**Ethernet**
VMware Network Adapter V..
S: 0 R: 0 Kbps**Ethernet**
VMware Network Adapter V..
S: 0 R: 0 Kbps**WiFi**
Wi-Fi
S: 0 R: 0 Kbps**GPU 0**
Intel(R) HD Graphics 530
16%**GPU 1**
NVIDIA GeForce GTX 960M

Memory

Memory usage



60 seconds

Memory composition



In use (Compressed)	Available	Speed:	1600 MHz
9.9 GB (0 MB)	5.9 GB	Slots used:	2 of 2
Committed	Cached	Form factor:	SODIMM
10.3/20.9 GB	5.9 GB	Hardware reserved:	134 MB
Paged pool	Non-paged pool		
552 MB	383 MB		

Run new task



16.0 GB DDR3

15.9 GB



Processes



Performance



App history



Startup apps



Users



Details



Services



Settings

Performance



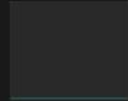
CPU
20% 3.00 GHz



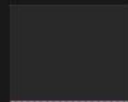
Memory
10.0/15.9 GB (63%)



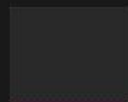
Disk 0 (C:)
SSD
2%



Disk 1 (D: P: E: S: H: G:)
HDD
0%



Ethernet
VMware Network Adapter V..
S: 0 R: 0 Kbps



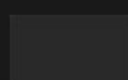
Ethernet
VMware Network Adapter V..
S: 0 R: 0 Kbps



WiFi
Wi-Fi
S: 0 R: 0 Kbps



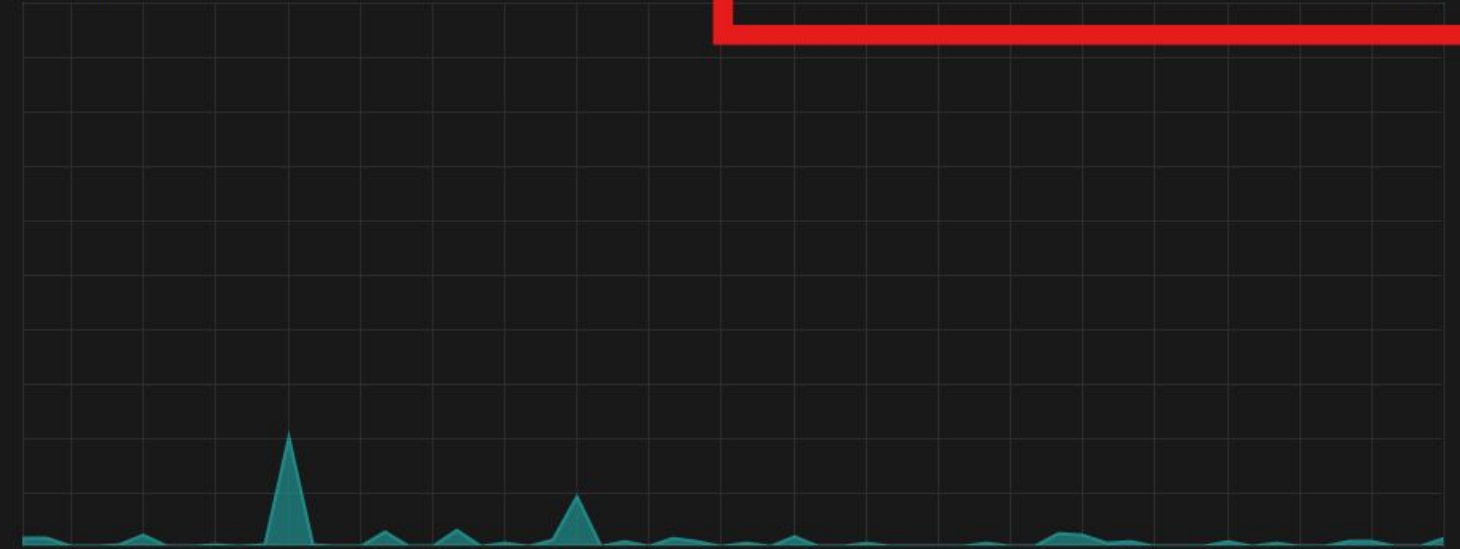
GPU 0
Intel(R) HD Graphics 530
2%



GPU 1
NVIDIA GeForce GTX 960M

Disk 0 (C:)

Active time



60 seconds

Disk transfer rate



60 seconds

Active time	Average response time	Capacity:	239 GB
2%	1.1 ms	Formatted:	239 GB
Read speed	Write speed	System disk:	Yes
0 KB/s	45.0 KB/s	Page file:	Yes
		Type:	SSD



Run new task



SAMSUNG MZNLN256HCHP-000H1

100%



Processes

Performance

App history

Startup apps

Users

Details

Services

Settings

Performance

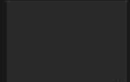
Run new task



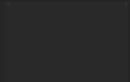
CPU
96% 3.09 GHz



Memory
9.9/15.9 GB (62%)



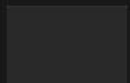
Disk 0 (C:)
SSD
7%



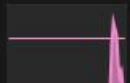
Disk 1 (D: P: E: S: H: G:)
HDD
0%



Ethernet
VMware Network Adapter V..
S: 0 R: 0 Kbps



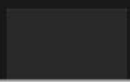
Ethernet
VMware Network Adapter V..
S: 0 R: 0 Kbps



WiFi
Wi-Fi
S: 0.2 R: 1.1 Mbps



GPU 0
Intel(R) HD Graphics 530
17%



GPU 1
NVIDIA GeForce GTX 960M

Disk 1 (D: P: E: S: H: G:)

Active time



60 seconds

Disk transfer rate

100 KB/s

0

Active time	Average response time	Capacity:	932 GB
0%	0 ms	Formatted:	932 GB
Read speed	Write speed	System disk:	No
0 KB/s	0 KB/s	Page file:	No
		Type:	HDD

WDC WD10SPZX-08Z10

100%



Processes



Performance



App history



Startup apps



Users



Details

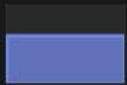
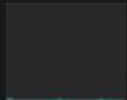
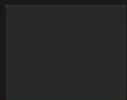
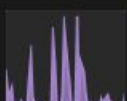
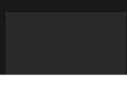


Services



Settings

Performance

**CPU**
28% 3.12 GHz**Memory**
9.9/15.9 GB (62%)**Disk 0 (C:)**
SSD
1%**Disk 1 (D: P: E: S: H: G:)**
HDD
0%**Ethernet**
VMware Network Adapter V..
S: 0 R: 0 Kbps**Ethernet**
VMware Network Adapter V..
S: 0 R: 0 Kbps**WiFi**
Wi-Fi
S: 0 R: 0 Kbps**GPU 0**
Intel(R) HD Graphics 530
14%**GPU 1**
NVIDIA GeForce GTX 960M

Run new task



Ethernet

VMware Virtual Ethernet Adapter for VMnet1

Throughput

100 Kbps

60 seconds

Send

0 Kbps

Receive

0 Kbps

Adapter name: VMware Network Adapter VMnet1

Connection type: Ethernet

IPv4 address: 192.168.52.1

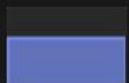
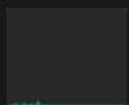
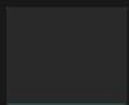
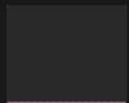
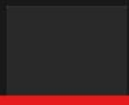
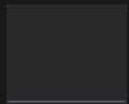
IPv6 address: fe80::4d3f:f039:e98a:c5b0%15



Performance

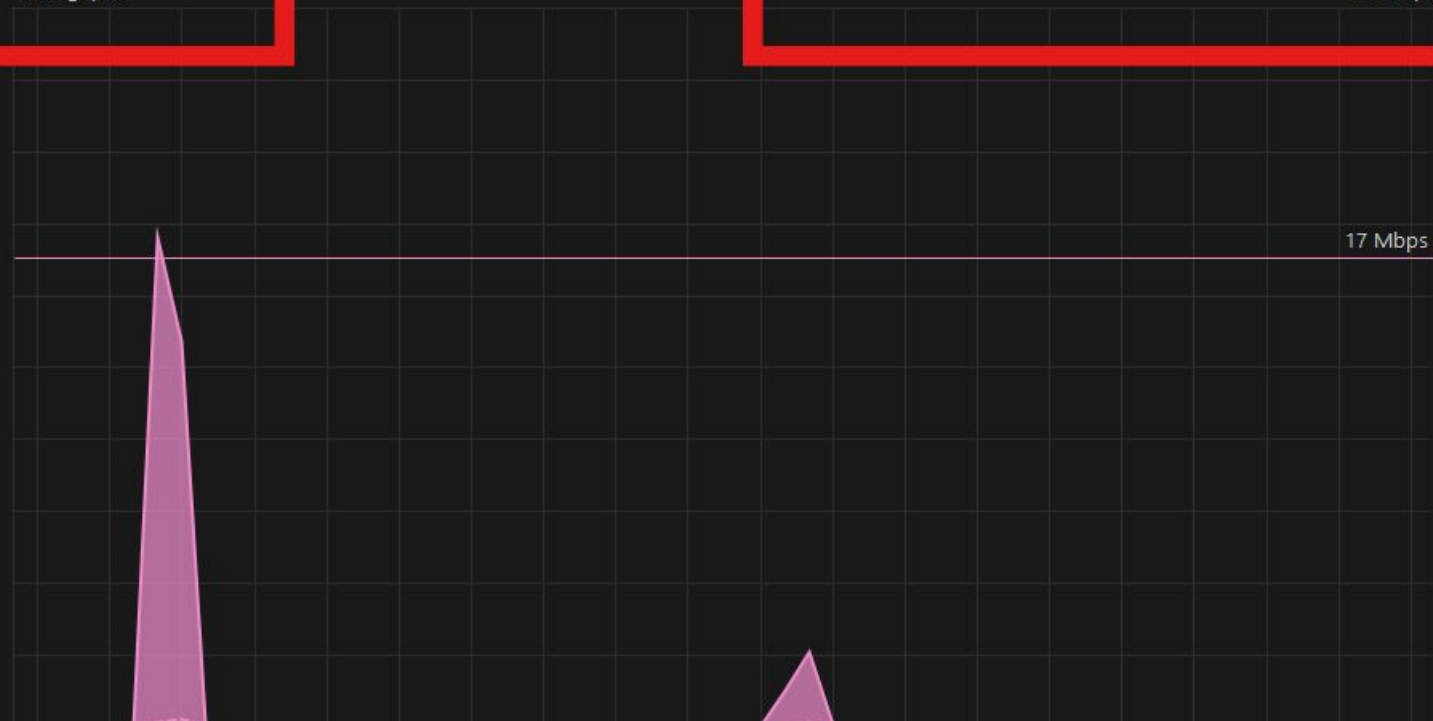


23% 3.11 GHz

**Memory**
9.8/15.9 GB (62%)**Disk 0 (C:)**
SSD
0%**Disk 1 (D: P: E: S: H: G:)**
HDD
0%**Ethernet**
VMware Network Adapter V..
S: 0 R: 0 Kbps**Ethernet**
VMware Network Adapter V..
S: 0 R: 0 Kbps**WiFi**
Wi-Fi
S: 0 R: 0 Kbps**GPU 0**
Intel(R) HD Graphics 530
8%**GPU 1**
NVIDIA GeForce GTX 960M
0%

WiFi

Throughput



00 seconds

0

Send

0 Kbps

Receive

0 Kbps

Adapter name:

Wi-Fi

SSID:

ptcl flash fiber.5G

Connection type:

802.11ac

IPv4 address:

192.168.1.6

IPv6 address:

fe80::1a43:e101:9933:c534%4

Signal strength:



Run new task



Intel(R) Dual Band Wireless-AC 3165

26 Mbps

17 Mbps



Processes



Performance



App history



Startup apps



Users



Details



Services



Settings

Performance

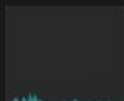


33% 3.03 GHz



Memory

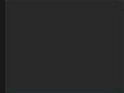
10.1/15.9 GB (64%)



Disk 0 (C:)

SSD

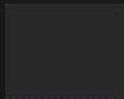
1%



Disk 1 (D: P: E: S: H: G:)

HDD

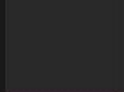
0%



Ethernet

VMware Network Adapter V..

S: 0 R: 0 Kbps



Ethernet

VMware Network Adapter V..

S: 0 R: 0 Kbps



WiFi

Wi-Fi

S: 0 R: 0 Kbps



GPU 0

Intel(R) HD Graphics 530

2%



GPU 1

NVIDIA GeForce GTX 960M

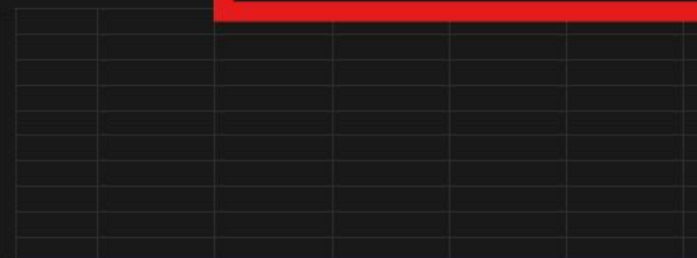
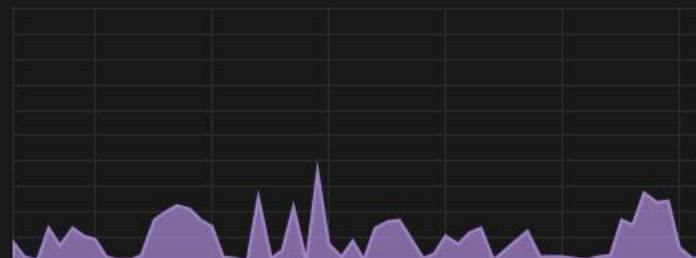
0%

GPU

3D

2%

Copy

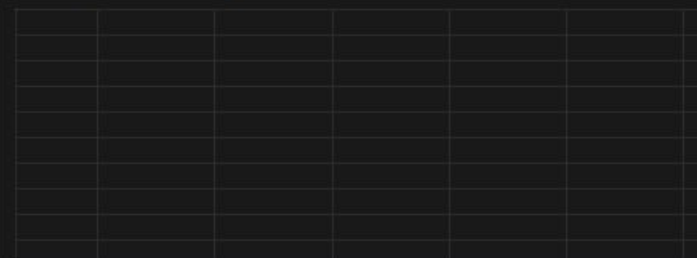
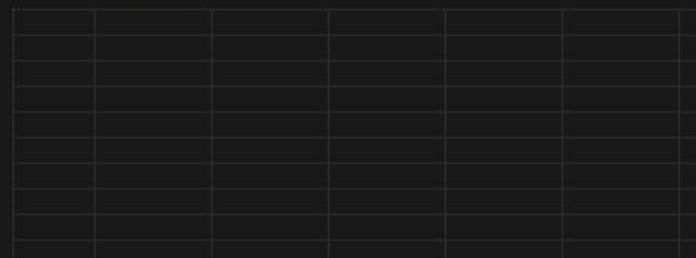


Video Decode

0%

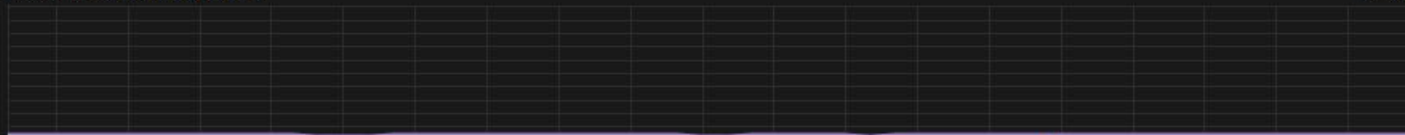
Video Processing

0%



Shared GPU memory usage

7.9 GB



Utilisation

2%

Shared GPU memory

0.3/7.9 GB

Driver version:

31.0.101.2111

Driver date:

7/19/2022

DirectX version:

12 (FL 12.1)

Physical location:

PCI bus 0, device 2, function 0

GPU Memory

0.3/7.9 GB



Processes



Performance



App history



Startup apps



Users



Details



Services



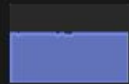
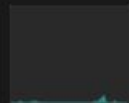
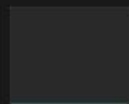
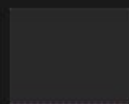
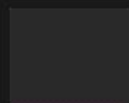
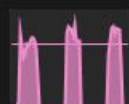
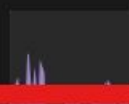
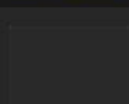
Settings

Performance

Run new task



33% 3.06 GHz

**Memory**
10.0/15.9 GB (63%)**Disk 0 (C:)**
SSD
0%**Disk 1 (D: P: E: S: H: G:)**
HDD
0%**Ethernet**
VMware Network Adapter V..
S: 0 R: 0 Kbps**Ethernet**
VMware Network Adapter V..
S: 0 R: 0 Kbps**WiFi**
Wi-Fi
S: 0 R: 0 Kbps**GPU 0**
Intel(R) HD Graphics 530**GPU 1**
NVIDIA GeForce GTX 960M
0%

GPU

3D

0%

Copy

0%

Video Encode

0%

Video Decode

0%

Dedicated GPU memory usage

4.0 GB

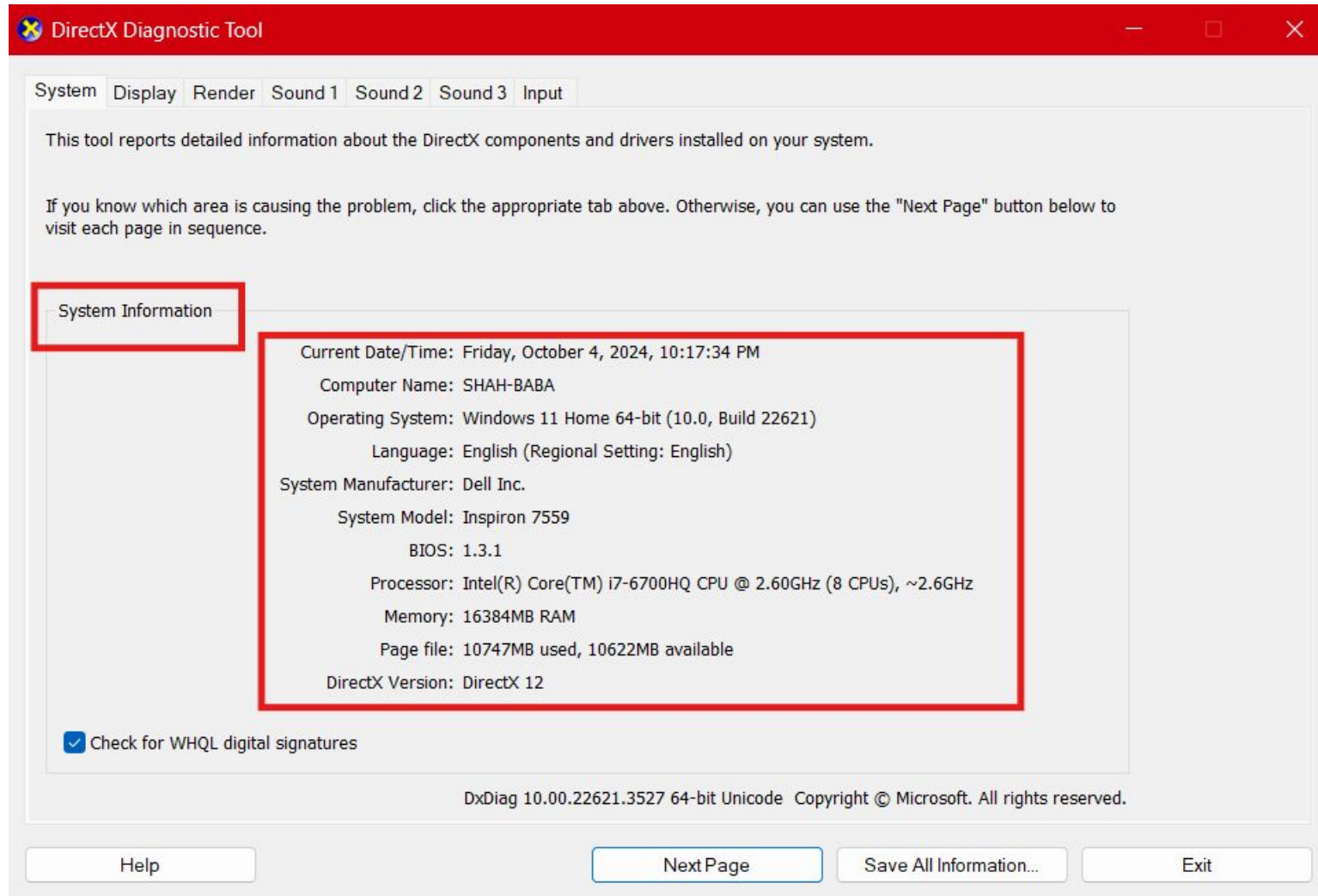
Shared GPU memory usage

7.9 GB

Utilisation	Dedicated GPU memory	Driver version:	32.0.15.6109
0%	0.0/4.0 GB	Driver date:	9/5/2024
GPU Memory	Shared GPU memory	DirectX version:	12 (FL 11.0)
0.0/11.9 GB	0.0/7.9 GB	Physical location:	PCI bus 2, device 0, function 0
		Hardware reserved memory:	48.8 MB

NVIDIA GeForce GTX 960M

Search DXDIAG



System **Display** Render Sound 1 Sound 2 Sound 3 Input

Device

Name: Intel(R) HD Graphics 530
Manufacturer: Intel Corporation
Chip Type: Intel(R) HD Graphics Family
DAC Type: Internal
Device Type: Full Display Device
Approx. Total Memory: 8253 MB
Display Memory: 128 MB
Shared Memory: 8125 MB

DirectX Features

DirectDraw Acceleration: Enabled
Direct3D Acceleration: Enabled
AGP Texture Acceleration: Enabled
DirectX 12 Ultimate: Disabled

Notes

- No problems found.

Drivers

Main Driver: igdumdim64.dll,igd10iumd64.dll,igd10c
Version: 31.0.101.2111
Date: 7/19/2022 05:00:00
WHQL Logo'd: Yes
Direct3D DDI: 12
Feature Levels: 12_1,12_0,11_1,11_0,10_1,10_0,9_3
Driver Model: WDDM 2.1

Help

Next Page

Save All Information...

Exit



System Display **Render** Sound 1 Sound 2 Sound 3 Input

Device

Name: NVIDIA GeForce GTX 960M
Manufacturer: NVIDIA
Chip Type: NVIDIA GeForce GTX 960M
DAC Type: Integrated RAMDAC
Device Type: Render-Only Display Device
Approx. Total Memory: 12172 MB
Display Memory 4047 MB
Shared Memory: 8125 MB

DirectX Features

DirectDraw Acceleration: Enabled
Direct3D Acceleration: Enabled
AGP Texture Acceleration: Enabled
DirectX 12 Ultimate: Disabled

Drivers

Main Driver: nvlumdx.dll,nvlumdx.dll,nvlumdx.d
Version: 32.0.15.6109
Date: 9/5/2024 05:00:00
WHQL Logo'd: Yes
Direct3D DDI: 12
Feature Levels: 11_0,10_1,10_0,9_3,9_2,9_1
Driver Model: WDDM 3.1

Notes

- No problems found.

Help

Next Page

Save All Information...

Exit

System Display Render **Sound 1** Sound 2 Sound 3 Input

Device

Name: Speakers / Headphones (Realtek Audio)
Hardware ID: HDAUDIO\FUNC_01&VEN_10EC&DEV_0256&SUBSYS_102
Manufacturer ID: N/A
Product ID: N/A
Type: N/A
Default Device: Yes

Drivers

Name: RTKVHD64.sys
Version: 6.0.1.8578 (English)
Date: 11/20/2018 05:00:00
WHQL Logo'd: Yes
Other Files:
Provider: Realtek Semiconductor Corp.

Notes

- No problems found.

Help

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Exit

System Display Render Sound 1 Sound 2 Sound 3 **Input**

DirectInput Devices

Device Name	Status	Controller ID	Vendor ID	Product ID	Force Feedback Driver
Mouse	Attached	n/a	n/a	n/a	n/a
Keyboard	Attached	n/a	n/a	n/a	n/a
HID I2C Device	Attached	0	0x04F3	0x3012	n/a
Airplane Mode Switch	Attached	0	0x413C	0xABCE	n/a
E-DRA EK361W	Attached	0	0x0C45	0x8006	n/a
F-DRA FK361W	Attached	0	0x0C45	0x8006	n/a

Input Related Devices

- + USB Devices
- + PS/2 Devices

Notes

- No problems found.

Help

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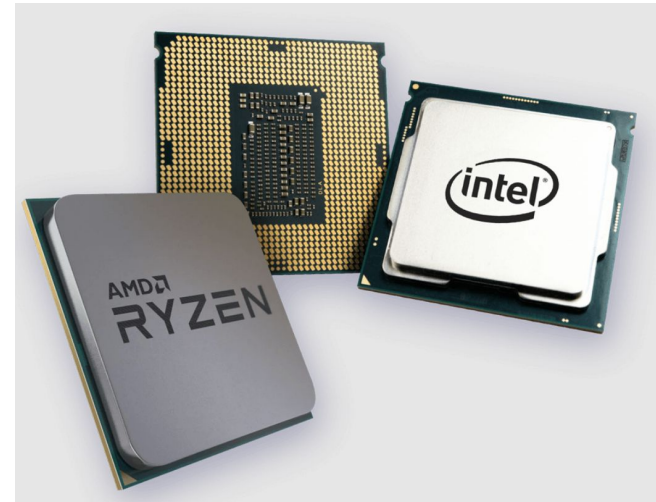
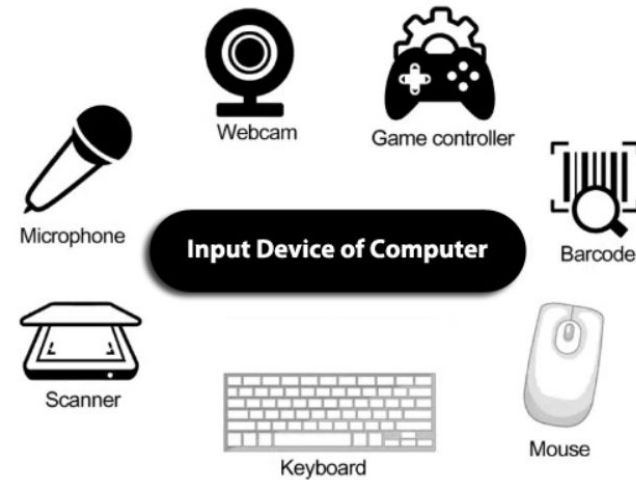
Introduction to Computers

- **Definition of a Computer:**

- A computer is an electronic device that can perform a wide variety of tasks by following programmed instructions.
- Its primary purpose is to process data into information.
- Computers can handle millions of instructions per second, making them powerful tools for automation, data analysis, and problem-solving.

Introduction to Computers

- **Input:** Data is provided by users via input devices like a keyboard, mouse, or scanner.
- **Processing:** The CPU processes the data by following the program's instructions to perform calculations or decisions.



Introduction to Computers

- **Output:** Processed data is then presented to the user as information, through devices like monitors, printers, or speakers.
- **Storage:** Data and instructions are stored for future use on storage devices such as hard drives or SSDs.



Introduction to Computers

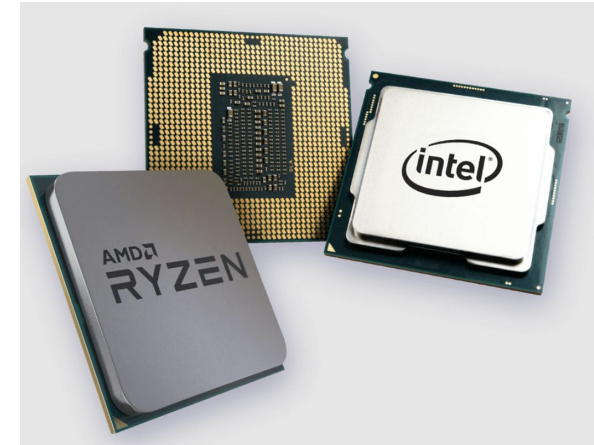
- **Types of Computers:**
- **Personal Computers (PCs):** Used by individuals for personal tasks like writing, browsing, or gaming. These include desktops and laptops.
- **Servers:** Powerful computers that manage network resources and services, such as websites or databases.
- **Mobile Devices:** Smartphones and tablets, which have computing capabilities but are portable and optimized for mobility.
- **Embedded Systems:** Computers integrated into other devices like cars, appliances, and medical equipment, designed to perform specific control tasks.



Identify Computer Components (Opening a PC and Identifying Parts)

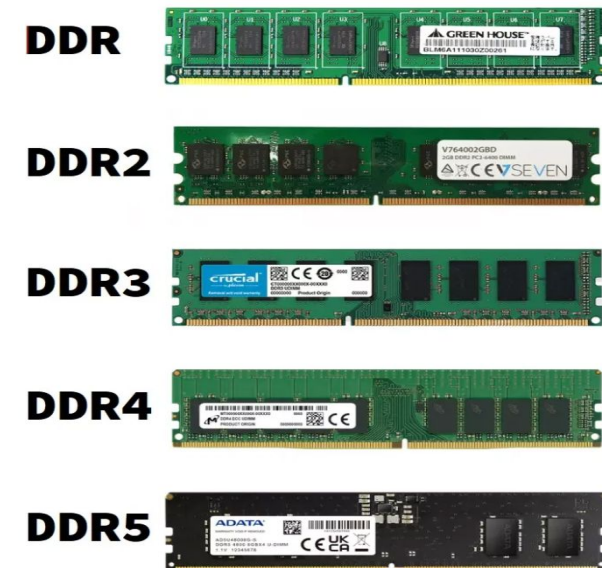
- **Central Processing Unit (CPU):**

- Known as the "brain" of the computer.
- It performs all calculations and executes instructions.
- CPUs have multiple cores, allowing them to handle many tasks simultaneously.
- Key brands: Intel, AMD Ryzen.



- **Random Access Memory (RAM):**

- Temporary storage for active data and processes.
- The more RAM, the better the system's ability to handle multiple tasks at once (multitasking).
- RAM is volatile, meaning data is lost when the power is off.



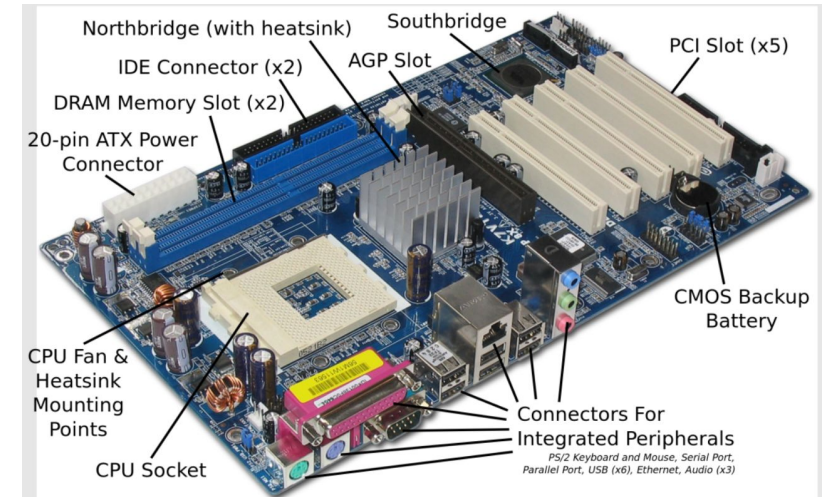
Identify Computer Components (Opening a PC and Identifying Parts)

- **Motherboard:**

- The main circuit board connecting all computer components.
- Hosts the CPU, RAM, and expansion slots for additional components like the GPU.
- It manages communication between various parts of the computer.

- **Power Supply Unit (PSU):**

- Converts electrical energy from the power outlet into usable energy for computer components.
- Provides different voltage levels for different components.
- Rated in watts (higher wattage means it can support more powerful systems).



Identify Computer Components (Opening a PC and Identifying Parts)

- **Graphics Processing Unit (GPU):**

- Handles rendering of images, video, and graphics.
- Essential for gaming, video editing, and tasks that require graphical computations (e.g., CAD).
- Can be integrated into the motherboard or installed as a dedicated component.



- **Cooling System (Fans, Heatsinks):**

- Prevents overheating by dissipating heat generated by the CPU, GPU, and other components.
- Fans draw cool air into the case and expel hot air, while heatsinks absorb and distribute heat.



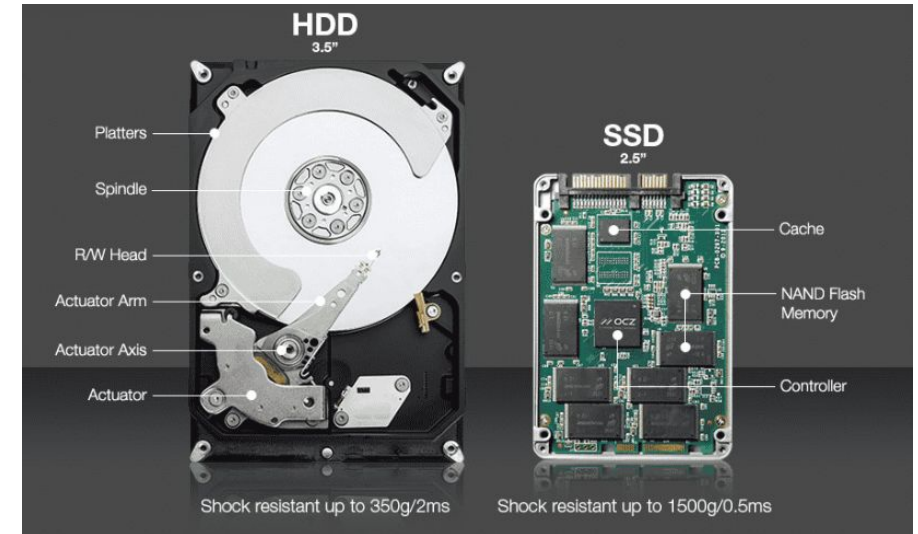
Identify Computer Components (Opening a PC and Identifying Parts)

- **Hard Drive / Solid State Drive (HDD/SSD):**

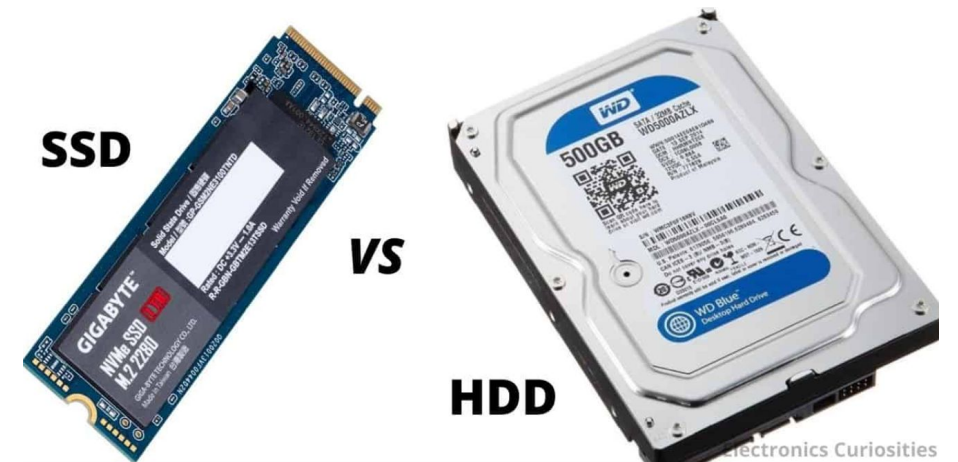
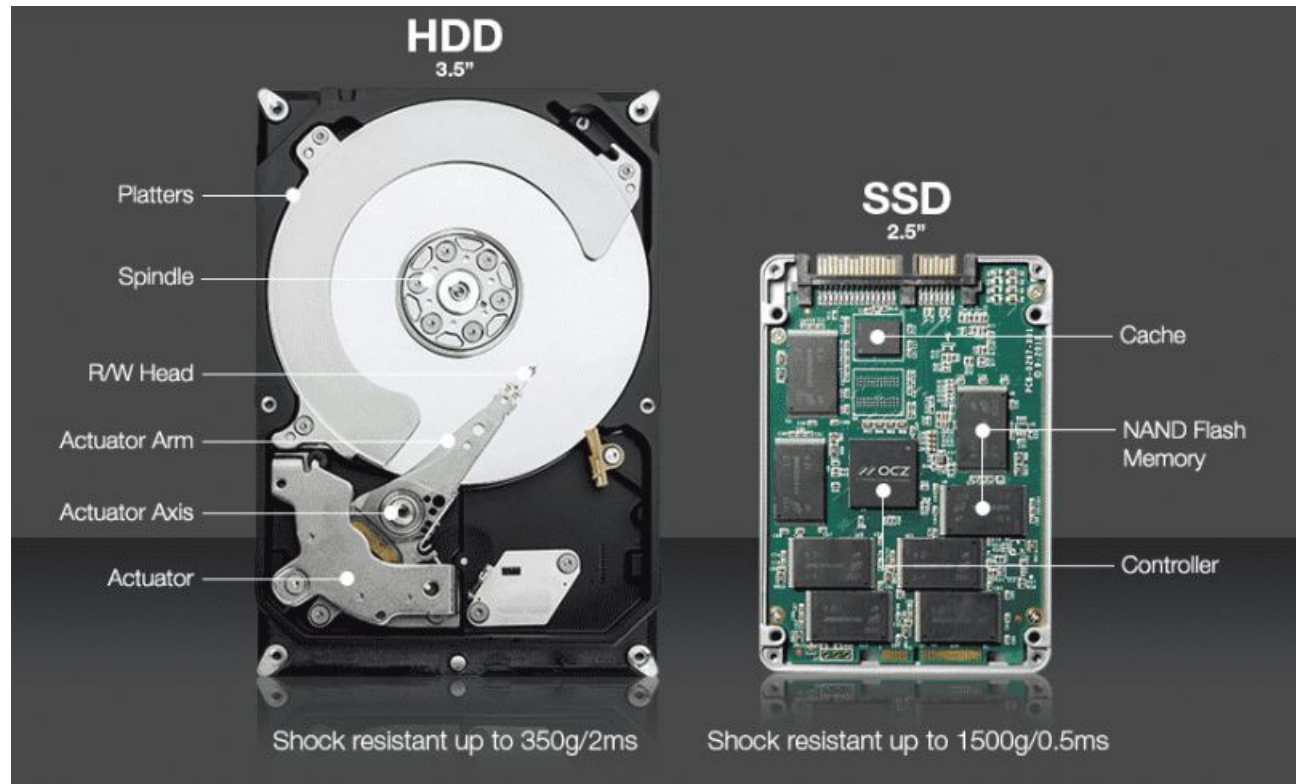
- HDD: Magnetic storage with spinning disks; offers high capacity but slower access speeds.
- SSD: Flash-based storage; much faster than HDDs but generally more expensive.

- **Optical Drive (if applicable):**

- Reads and writes data on CDs, DVDs, or Blu-ray discs.
- Becoming less common with the rise of digital downloads and cloud storage.



HDD Vs SSD



Explore Storage Devices and Install a Hard Drive

- **Types of Storage Devices (HDD, SSD, USB Flash Drive):**

- **HDD (Hard Disk Drive):** Traditional mechanical storage with large capacity (up to several TB) but slower speeds (in MB/s). Common for large-scale data storage.
- **SSD (Solid State Drive):** Faster and more reliable than HDD, SSDs use flash memory. They're more expensive but can greatly improve system performance, especially in boot times and data retrieval.
- **USB Flash Drive:** A portable, small-capacity (up to several GB) storage device used for file transfers and backups.

Internal vs External Storage:

- **Internal Storage:** Storage that is built into the computer system (HDD/SSD). It's typically the primary storage for the operating system, programs, and files.
- **External Storage:** External drives like USB, external HDD, or SSDs can be connected via ports (e.g., USB or Thunderbolt) and used for backups or transferring large files.

Capacity and Speed Considerations:

- **Capacity:** Measured in gigabytes (GB) or terabytes (TB). More capacity allows for storing larger amounts of data, such as media files, applications, or games.
- **Speed:** SSDs are typically measured in MB/s or GB/s and offer much faster read/write speeds than HDDs, which improves overall performance (faster boot times, quicker file access).

Steps to Install a Hard Drive:

- **Shut down and unplug the PC:** Ensure that no power is running through the system to avoid short circuits or damage to components.
- **Open the case:** Use a screwdriver (if necessary) to open the side panel of the computer case to access internal components.
- **Locate available drive bay:** Most desktop cases have slots or bays for installing hard drives or SSDs. Identify a free slot where the drive can be installed.
- **Connect the SATA cable and power connector:** SATA cables are used to connect the hard drive to the motherboard for data transfer. Connect the power cable from the PSU to the drive.
- **Secure the drive in place:** Use screws or clips to secure the drive to the case so it doesn't move during operation.
- **Close the case and start the PC:** After securing the drive, close the case, plug the computer back in, and power it up. Check that the drive is recognized by the operating system under storage management.